Beyond the edge of stability: periodic windows, quasi-periodicity and phase locking in the learning-rate plane of gradient descent

Authorship to be finalized by the human researcher

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Abstract

Full-batch gradient descent with a large learning rate rarely converges to a minimum: it oscillates at the edge of stability, cycles, or behaves chaotically. Maps of trainability over learning-rate planes summarize this as a binary label, convergent or divergent, and so do not say which of these behaviors occurs where. We resolve the dynamics behind such maps in two small tanh networks trained with separate learning rates for their two layers.

(1) In a prospectively designed and independently audited study, the learning-rate planes contain 61 numerically certified attracting periodic orbits of the full parameter-space update map, with periods from 4 to 58, whose domains are surrounded mainly by rerun-confirmed chaos. Across six coarse maps, 87.1% to 99.97% of the cells that a published binary convergence score labels convergent are not fixed points after 20,000 updates.

(2) In exploratory analyses, the windows follow a quasi-periodic route to chaos. On all 35 traced slices whose starting point is the edge-of-stability period-2 orbit, that orbit loses stability through a Neimark–Sacker bifurcation of the two-step map. On one slice the crossing multiplier predicts a rotation number of 0.2180, and the first quasi-periodic cell rotates at 0.2180. Of the 56 testable windows, 35 are well-ordered circle orbits with rotation m/q and period $2q$ or a doubling of it, against 3.58 expected for random angles. Only 2 of 61 windows meet draft criteria for classical shrimp anatomy.

Takeaway: in these networks, the region that a binary map calls trainable holds a structured transition from oscillation to chaos, including quasi-periodic training and phase-locked windows. The mechanism is exploratory and still needs a test on fresh networks and seeds; nothing here concerns stochastic or large-scale training or establishes a training benefit.

Study status. The map study (Section 4) was designed and pushed before any outcome and checked by an independent audit; the mechanism analyses (Section 5) are exploratory. Provenance for every figure and quoted number is described in Section 9.

1 Introduction

Training with a learning rate beyond the classical stability limit is common practice. At the edge of stability, the sharpness of the loss hovers near $2/\eta$ while the loss still decreases over long horizons [5], and simplified models show period doubling and chaos as the step size grows [3, 11, 16]. Over

two-dimensional hyperparameter planes, the boundary between trainable and divergent settings is fractal [19, 20, 25, 29]. These maps record whether training stays bounded and settles, not what it does.

We ask what the dynamics inside such maps are. We treat full-batch gradient descent with separate learning rates for the hidden and output layers as a two-parameter family of maps on parameter space. For every cell of a learning-rate grid we classify the long-run behavior (fixed point, periodic orbit with its period, chaos, drift, divergence), certify periodic orbits with Newton's method on the exact Jacobian, and follow selected orbits and their stability boundaries through the plane.

Prior work. Periodic windows in one-dimensional learning-rate sweeps date back at least to van der Maas et al. [30]. Period-colored maps over optimizer hyperparameters exist without full-state verification [28]. Continuation of flip and catapult thresholds has been carried out for one-dimensional reductions of gradient descent [11], which also derives complex unstable multipliers of a 2-cycle in a two-mode linear network. Riddled basins and learning-rate-dependent reproducibility are documented by Ly and Gong [21]. Arnold tongues and Neimark–Sacker bifurcations are reported for learning dynamics in games [22], and heavy-ball momentum has cycles that rotate by $1/K$ [10].

This paper's contribution.

- A period-resolved description of the learning-rate plane of full-batch gradient descent in two small networks, with 61 certified attracting windows embedded in chaos, confirmed by an independent audit.
- A direct comparison with the published binary convergence score on the same cells, showing that most cells it calls convergent do not converge.
- Exploratory evidence that the windows are organized by a quasi-periodic route: Neimark–Sacker bifurcation of the period-2 orbit, invariant circles with measured rotation numbers, phase-locked orbits with period $2q$ at rotation m/q, and rotation-number staircases.
- Open code, receipts and a verification script that re-derives every figure and quoted number.

Not claimed.

- No claim about stochastic, minibatch or large-scale training, other architectures or real data (a real-data test is in progress; Section 5).
- No claim that periodic or quasi-periodic training helps or hurts generalization or speed.
- The mechanism in Section 5 is exploratory; it has not yet been tested on fresh networks and seeds.
- Classical shrimp organization is shown for a small minority of windows only.

2 Gradient descent as a family of maps

Let $L(\theta)$ be the full-batch training loss of a network with parameters $\theta \in \mathbb{R}^n$, split into hidden-layer and output-layer blocks. Gradient descent with per-layer learning rates is the map

$$\theta_{t+1} = F(\theta_t) = \theta_t - P\,\nabla L(\theta_t), \qquad P = \mathrm{diag}(\eta_h \mathbf{1}_{\mathrm{hidden}}, \eta_o \mathbf{1}_{\mathrm{output}}), \tag{1}$$

a family of maps indexed by the two rates (η_h, η_o). Its Jacobian is $DF(\theta) = I - PH(\theta)$ with H the Hessian of L. An orbit of period p is stable when all eigenvalues (multipliers) of $DF(\theta_{p-1}) \cdots DF(\theta_0)$ lie inside the unit circle, and it loses stability when a multiplier crosses the circle: at $+1$ (fold), at -1 (flip, which doubles the period) or as a complex pair $e^{\pm 2\pi i\omega}$ (Neimark–Sacker, which creates an invariant closed curve) [18].

Why the route starts at period 2. $DF(\theta)$ is similar to the symmetric matrix $I - P^{1/2}HP^{1/2}$, so at a fixed point all multipliers are real and a fixed point can only lose stability by fold or flip. The edge-of-stability oscillation is the period-2 orbit born at that flip. For a period-2 orbit $\{\theta_a, \theta_b\}$, the two-step Jacobian is similar to S_bS_a with $S_j = I - P^{1/2}H(\theta_j)P^{1/2}$. A product of two symmetric matrices has real eigenvalues when one factor is positive definite, but at the edge of stability each S_j has a negative eigenvalue in the oscillating direction, so complex multipliers, and with them a Neimark–Sacker bifurcation of the two-step map, become possible. (This elementary argument is ours; the same observation appears in a literature check summarized in Section 8.)

Circle maps and phase locking. On an invariant circle the dynamics has a rotation number ρ, the mean fraction of a turn per iterate. When $\rho = m/q$ is rational the motion can lock onto a periodic orbit; the locked parameter sets are Arnold tongues, rotation numbers along a parameter path form a devil's staircase, and the widest plateau between two locked rationals is typically their Farey mediant [1, 12, 13]. Breakdown of the circle leads to chaos [2, 7, 23, 24]. Because our circle belongs to the two-step map, a locked orbit at rotation m/q has period $2q$ for F.

Shrimps. In two-parameter families of dissipative maps, periodic windows inside chaos often take a characteristic "shrimp" shape bounded by fold and flip curves [8, 9, 15, 27]. We test for this anatomy explicitly rather than assume it.

3 Methods

Networks and data. S1 is the published mean-field tanh configuration of the fractal-trainability study [25]: width 8, inputs of dimension 8, no biases, as many standard-normal training points as parameters, mean squared error. S1R is S1 with weight decay 10^{-3}. S2 is a 13-parameter tanh regressor (two inputs, three hidden units, biases, weight decay 10^{-3}) fitted to 24 points from a random teacher. We refer to a network and seed together, for example S2-28011. Training is deterministic full-batch gradient descent in float64.

Maps. Each map is a 256×256 grid of logarithmically spaced rate pairs (10 maps of 65,536 cells, coarse and zoomed). Every cell runs 20,000 updates from the same initial parameters. The last 384 updates are screened for the smallest full-state recurrence period up to 64; the largest finite-time Lyapunov exponent (FTLE) is estimated with one tangent vector; cells are labeled fixed point, periodic, chaotic (FTLE above 0.005 per update), drift (parameter norm still growing), unresolved or divergent. The published binary score marks a run convergent when its normalized loss, averaged over the last 20 of the first 1,000 updates, is below one [25].

Certification and embedded windows. A connected component of equal-period cells qualifies as an embedded window when its period is at least 3, no cell of its exterior ring is a fixed point or has a proper divisor period, and at least half of the ring is chaotic on reruns. Its representative orbit is refined by damped Newton iteration on the exact Jacobian and must close, be primitive and

be attracting with margin 10^{-7}. The design, thresholds and analysis were pushed before outcomes were seen, and an independent audit repeated the checks.

Continuation and boundary type. From the center of a window, the certified orbit is continued along 64 rays in the rate plane until it loses stability, located by bisection, and the boundary is typed by the crossing multiplier: fold ($+1$), flip (-1) or Neimark–Sacker (complex pair). The same procedure traces where the period-2 orbit loses stability along slices of fixed output rate.

Rotation numbers and well-ordering. For a non-periodic cell we take the two-step samples $z_k = \theta_{2k}$, project them on their first two principal axes, and define the rotation number as the mean unwrapped angle increment in turns per two steps. For a certified orbit with q two-step points we sort the points by angle and call the orbit well-ordered (a Birkhoff orbit of rotation m/q) when every step advances the angular rank by the same shift m with $\gcd(m, q) = 1$. Orbits of 2 or 3 points pass trivially; only tests on at least 5 points count as evidence. The null is the pass rate of the same test on uniformly random angles. A refined version also orders points along the shortest closed tour through them and tests the points of $F^{2^{k+1}}$ ($k = 1, 2, 3$), which detects orbits born by period doubling inside a tongue; its null sums the per-test rates.

4 Periodic windows inside chaotic training

Evidence label: PROSPECTIVELY FROZEN, AUDITED.

The learning-rate planes of S1R and S2 contain 61 embedded windows that qualify on both the producer's and the auditor's evidence (62 and 63 respectively; the disagreements all sit at the 50% ring threshold). All 154 attempted orbit certifications verified. Confirmed periods range from 4 to 58 and include the odd periods 5, 11, 13, 17 and 29. Their exterior rings were checked with 2,801 reruns, and 157 of 160 randomly chosen chaotic cells stayed chaotic on a rerun with a different tangent vector.

Figure 1 compares these dynamics with the published binary score. In every coarse map, most of the cells the score calls convergent are not fixed points: 87.1% to 99.97%. The published configuration without weight decay (S1) has 0 periodic cells in both seeds: its runs diverge, stay chaotic, or drift toward large weights while the loss slowly falls. Adding weight decay makes the windows appear.

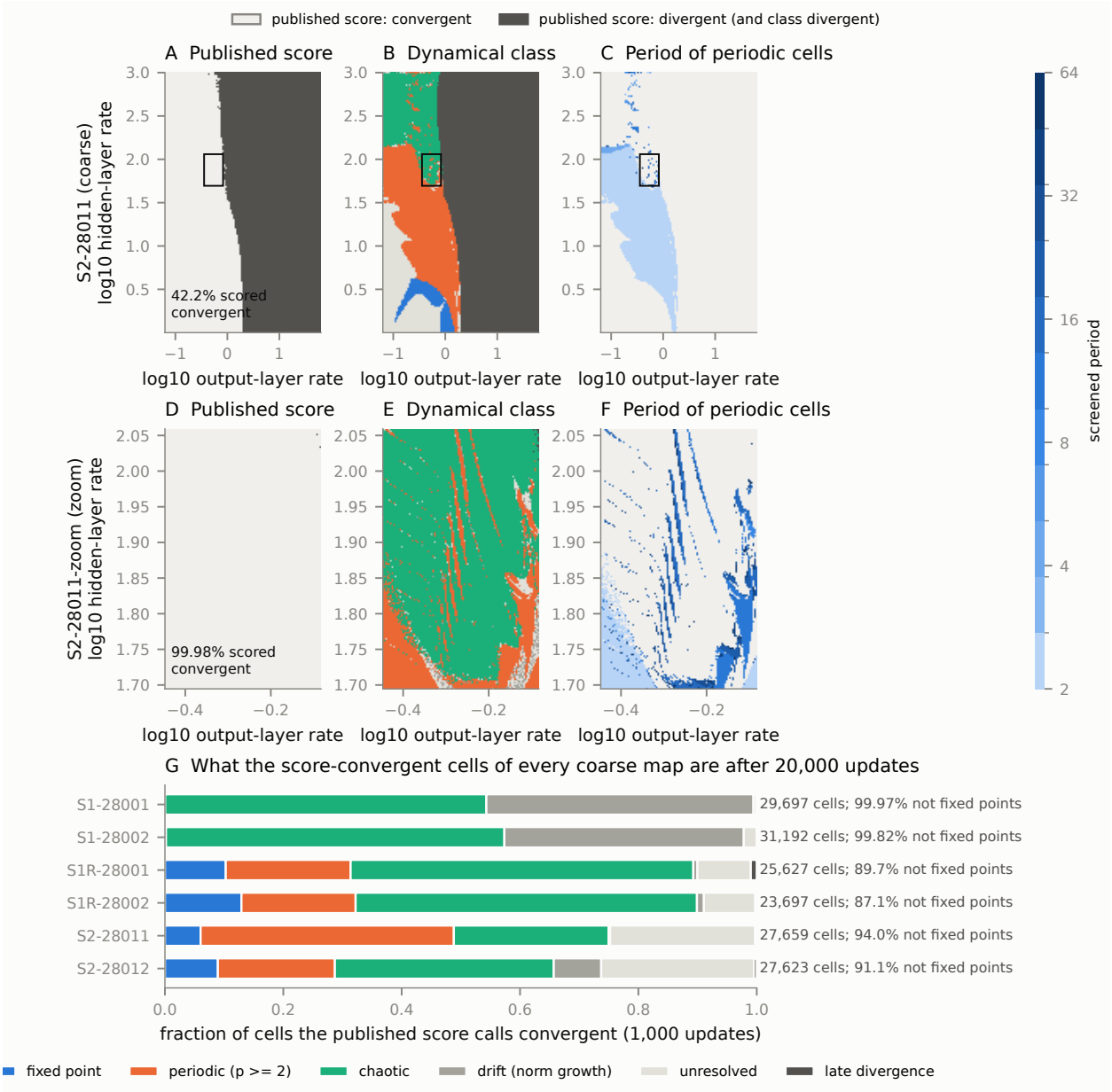


Figure 1: The published binary score calls almost everything in these planes convergent, while the same cells hold fixed points, periodic orbits of many periods, chaos and drift. Panels A to F: one network (S2-28011), coarse and zoomed maps; the rectangle marks the zoom. Panel G: composition of the cells the score calls convergent, for all six coarse maps; only the blue segment converges to a fixed point. The comparison is cell by cell on the same runs; the score uses the first 1,000 updates and the classes use 20,000. Labels other than certified orbits are finite-time screening labels. *Generated by* `paper/gd-route/figures/f1_binary_vs_period.py`; *receipt* `figures/f1_binary_vs_period.provenance.json`.

5 An exploratory quasi-periodic route

Evidence label: EXPLORATORY. The analyses in this section were chosen while exploring and used data that shaped the hypothesis.

5.1 The period-2 orbit loses stability through a Neimark–Sacker bifurcation

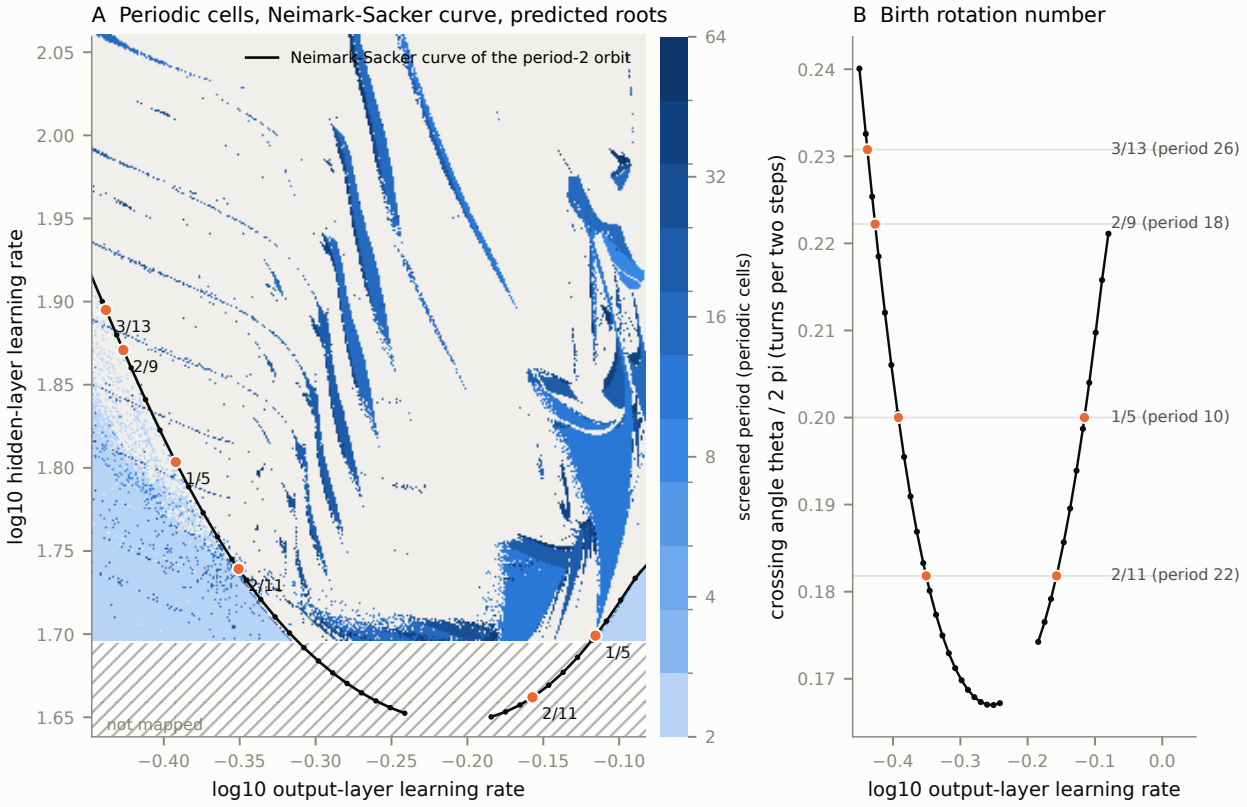


Figure 2: The edge-of-stability period-2 orbit loses stability through a complex pair of multipliers along the whole traced curve. A: periodic cells of a 1,024 × 1,024 map (shade: period) with the traced Neimark–Sacker curve (black) and the points where its crossing angle equals a rational m/q with $q \leq 13$ (orange). B: crossing angle along the curve. On 5 of 40 slices the starting point was not on the period-2 orbit; those slices break the curve. Predicted tongue roots are interpolated between traced slices, and tongues near their roots are narrower than one map cell, so the map alone cannot confirm them. *Generated by* `paper/gd-route/figures/f2_ns_tongue_roots.py`; *receipt* `figures/f2_ns_tongue_roots.provenance.json`.

We continued the period-2 orbit of S2-28011 toward larger hidden-layer rates along 40 slices of fixed output rate. On all 35 slices that started on it, the orbit loses stability with a complex pair of two-step multipliers crossing the unit circle (Figure 2). The curve is U-shaped, with its lowest point at $\log_{10} \eta_h = 1.650$, and the crossing angle ω ranges from 0.1670 to 0.2401 turns per two steps, passing 2/11, 1/5, 2/9 and 3/13.

The crossing angle predicts the rotation of the training that follows. On the slice through the period-18 window, the curve interpolated to that slice places the bifurcation at $\log_{10} \eta_h = 1.738$ with $\omega = 0.2180$; the first quasi-periodic cell of an independent staircase computation lies at 1.738 and rotates at 0.2180.

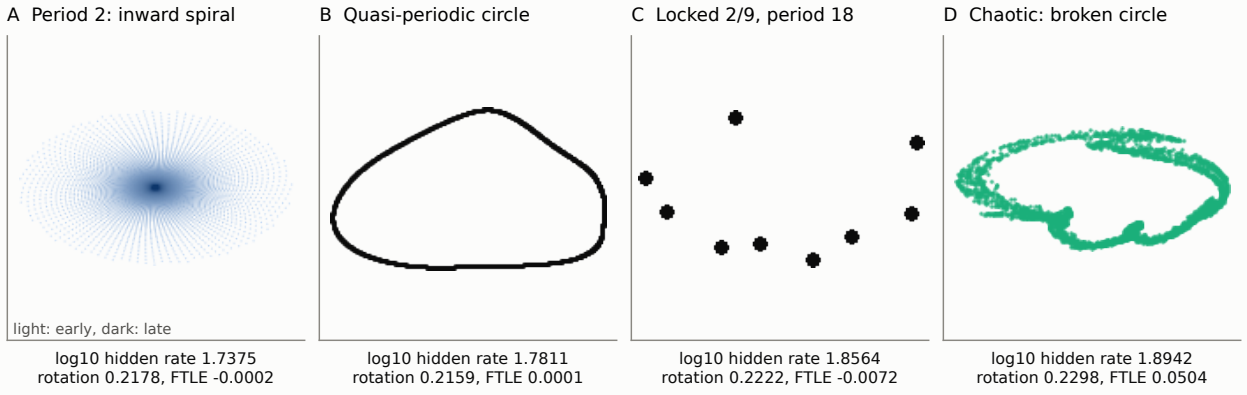


Figure 3: Four stages of the route along one slice: an inward spiral to the 2-cycle, a closed invariant curve, nine locked points, and a broken curve. Each panel shows every second parameter vector of one run projected on its own two principal axes. The rates are chosen by a fixed rule from the staircase receipt (the period-2 cell closest to onset, the median quasi-periodic cell, the middle of the longest 2/9 plateau, the median chaotic cell between the 2/9 and 1/4 plateaus), not by eye. Projections onto two axes can hide structure in the remaining dimensions. *Generated by* `paper/gd-route/figures/f6_route_portraits.py`; *receipt* `figures/f6_route_portraits.provenance.json`.

Figure 3 shows what training looks like along that slice. Beyond the bifurcation, the two-step samples lie on a closed curve (rotation 0.2159, FTLE 0.0001); inside the period-18 window they are nine points (rotation 0.2222, FTLE -0.0072); beyond it the curve thickens and breaks (FTLE 0.0504).

Placeholder. Rotation-number map of the S2-28011 plane overlaid with the Neimark–Sacker curve, testing whether the rotation just beyond the curve equals the crossing angle everywhere along it (computation in progress).

5.2 Rotation numbers form staircases

Along the slice through the period-18 window of S2-28011, the rotation number rises from the bifurcation value, locks at 2/9 for $\log_{10}\eta_h$ from 1.833 to 1.879, and locks again at 1/4 before chaos (Figure 4). The labeled plateaus are 5/23, 7/32, 2/9 and 1/4 in S2 and 1/9, 2/17, 1/8, 3/23 and 2/15 in S1R. Consecutive plateaus are Farey neighbors ($|m_1q_2 - m_2q_1| = 1$) in 3 of 3 pairs in S2, against 49% per pair for randomly drawn fractions in the same range, and in 4 of 4 pairs in S1R, where the null is 100% because every fraction with a small enough denominator in that range appears. The Farey pattern is therefore weak evidence in S2 and uninformative in S1R.

5.3 Most windows are phase-locked circle orbits

Across the 61 windows, the 3,904 rays end at 1,316 folds, 1,012 flips and 1,395 Neimark–Sacker crossings (181 rays reached the end of their range or stayed unresolved; Figure 5A). Tongue edges are folds of the locked orbit on the circle, while flips and secondary Neimark–Sacker crossings are expected where a locked orbit loses stability off the circle.

Among all certified orbits with at least 5 two-step points, 65 of 115 are well-ordered in their principal plane, against 1.57 expected for random angles. For the windows themselves, 35 of 56 testable windows pass the refined test, against a multiplicity-corrected null of 3.58; 7 of them pass first at a period-doubled level, for example period-20 orbits that are well-ordered under four steps

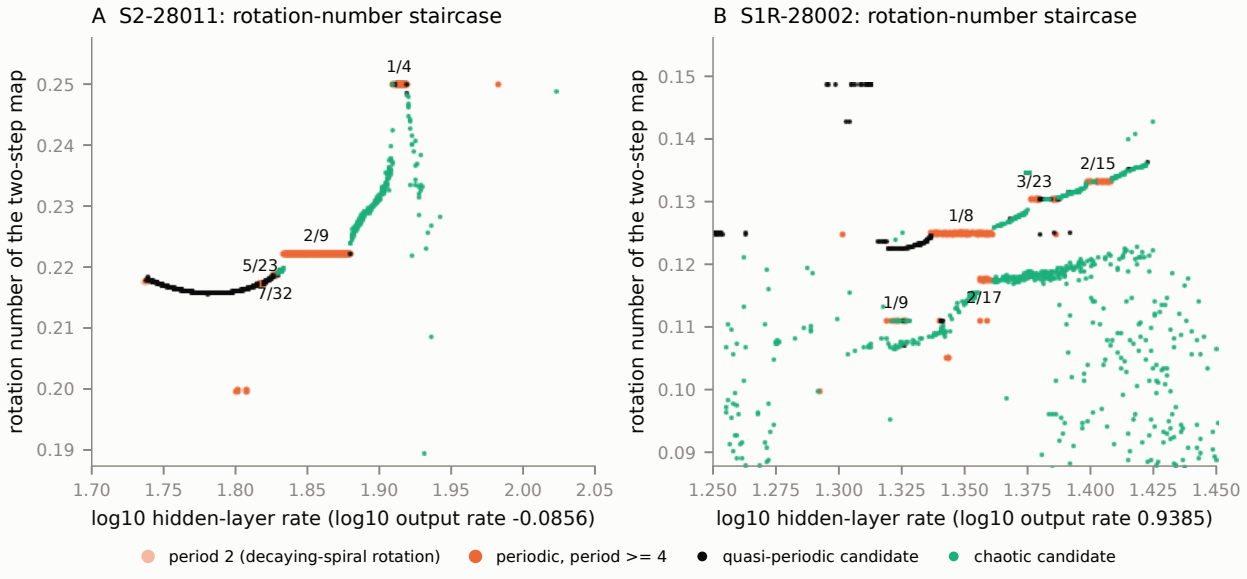


Figure 4: Along slices, quasi-periodic cells trace continuous rotation numbers and periodic cells sit on rational plateaus. Plateaus are detected automatically (locked cells with period $2q$ or a doubling of it, rotation within 6×10^{-4} of m/q) and labeled when they hold at least 8 cells. In S1R two rotation branches coexist over the same rates, a sign of coexisting attractors. Chaotic cells outside the plotted range are not shown. *Generated by* `paper/gd-route/figures/f3_staircases.py`; *receipt* `figures/f3_staircases.provenance.json`.

with rotation 2/5, twice 1/5. Every passing window has period $2q$ or $4q$ for its tongue denominator q (Figure 5C). Of the 5 odd-period windows, 0 are well-ordered (null 0.17), so they need a different explanation.

5.4 Classical shrimp anatomy is the exception

Draft criteria for classical shrimp anatomy require fold and flip boundaries each on at least 8 of 64 rays, at least half of the flip rays opening onto period $2p$ in a local map, and agreement between ray and map edges on at least three quarters of rays. 15 windows meet the boundary-count part and 2 meet all of it. The clearest, the period-18 window of S2-28011 (Figure 6), has 46 fold, 15 flip and 3 Neimark–Sacker rays; ray and map edges agree within two cells on 61 of 64 rays, and 10 of its flip rays open onto period 36. Neimark–Sacker boundaries, absent from classical shrimps, are common: 41 of 61 windows have at least 8 such rays.

Placeholder. Minimal two-parameter model (scalar factorization with weight decay and two learning rates), testing whether the same route appears where invariant circles can be drawn in the full state space (computation in progress).

Placeholder. Real-data network trained with a single learning rate, testing whether the period-2 orbit loses stability by Neimark–Sacker and whether loss spectra show the rotation (stream in progress).

Placeholder. Consequence for training: whether rotation number or locking predicts reproducibility or loss beyond the Lyapunov exponent and sharpness, against periodic and chaotic controls (stream in progress).

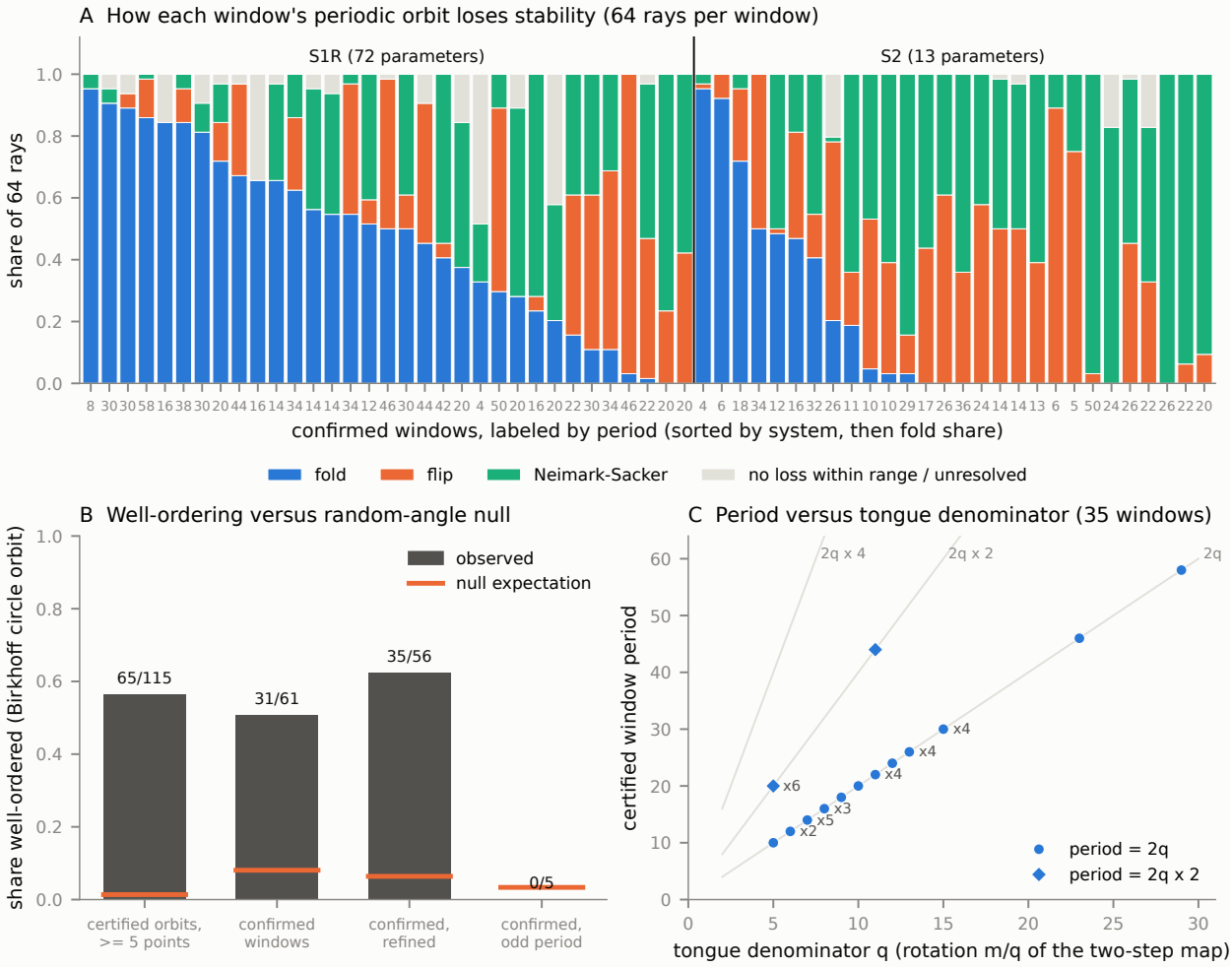


Figure 5: Across all 61 windows, stability boundaries mix fold, flip and Neimark–Sacker, and most testable windows are well-ordered circle orbits whose period is twice the tongue denominator or a doubling of it. A: boundary type along 64 rays per window. B: observed share of well-ordered orbits (gray) against the random-angle null (orange) on the same test. C: certified period against the tongue denominator for windows passing the refined test; the multiplicity of coincident points is printed. Orbits of 2 or 3 points pass trivially and are excluded from the refined share. *Generated by* `paper/gd-route/figures/f5_window_census.py`; *receipt* `figures/f5_window_census.provenance.json`.

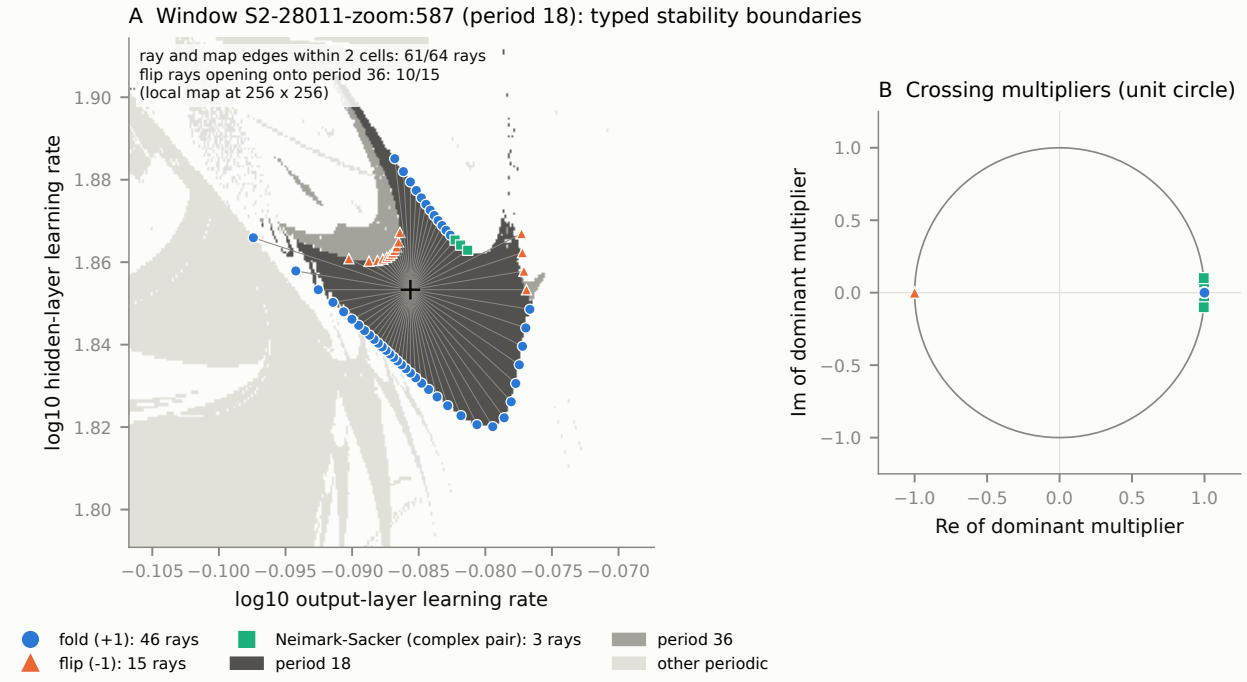


Figure 6: The period-18 window of S2-28011 has a fold-bounded body with flip edges opening onto period-36 bands. A: local map from the shared initial parameters with the window's orbit continued along 64 rays; markers show where and how it loses stability. B: crossing multipliers on the unit circle. This window was selected as the clearest example and is one of only 2 meeting the draft shrimp criteria; it is not representative (Figure 5). *Generated by* `paper/gd-route/figures/f4_window_anatomy.py`; *receipt* `figures/f4_window_anatomy.provenance.json`.

6 Discussion

Everything in this section is interpretation. The audited results in Section 4 stand on their own, and the hypotheses here need their own prespecified tests.

The picture that fits the exploratory evidence is the quasi-periodic route familiar from forced oscillators and circle maps [2, 7, 13]: the edge-of-stability oscillation is a period-2 orbit; past a Neimark–Sacker bifurcation of the two-step map, training moves on an invariant circle with a rotation number set by the crossing angle; rational rotations lock into Arnold tongues; and the circle breaks into chaos, with period doubling inside tongues along the way. The argument in Section 2 explains why this route needs at least period 2 in gradient descent, and why per-layer rates are not required for it in principle. Under this reading, the windows of Section 4 are mostly tongue windows, and classical shrimps are a special case.

Three observations do not fit yet. The odd-period windows are not circle orbits; a one-dimensional period-doubling route of the kind organized by shrimp hubs [15] is a candidate. Windows also coexist with other attractors, so maps started from one initial point show which attractor that point reaches rather than the full existence region of a window. And the evidence comes from two small networks.

A confirmatory test on fresh seeds and at least one new network would fix in advance: (a) a quasi-periodic band between the period-2 region and chaos, with rotation numbers that start at the crossing angle; (b) locked windows with period $2q$ at rotation m/q; (c) Neimark–Sacker loss of stability of the period-2 orbit at the band's onset; and (d) an explicit null for any ordering claim about plateaus.

7 Limitations

The networks have 13 and 72 parameters and are trained on synthetic data with deterministic full-batch updates in float64. Screening labels are finite-time; chaotic labels rest on a single tangent vector and reruns. Rotation numbers and well-ordering use two-dimensional projections of curves that live in 13 or 72 dimensions. The mechanism analyses were chosen after seeing related outcomes, and the staircase and Neimark–Sacker analyses cover two slices and one curve. No result addresses generalization, training speed, or modern training practice.

8 Related work

Edge-of-stability dynamics and their simplified models include Cohen et al. [5], Kalra et al. [16], Chen and Bruna [4], Song and Yun [26], Chen et al. [3], Cohen et al. [6] and Hofmann [11]; chaos in deterministic training is discussed by Kong and Tao [17] and Jiménez-González et al. [14]. Fractal trainability boundaries are studied by Sohl-Dickstein [25], Torkamandi [29], Liu [20] and Liang and Montúfar [19], and riddled basins and reproducibility by Ly and Gong [21]. Shrimp-shaped windows and their organization are reviewed in Gallas [8, 9], Jones [15], Stoop et al. [27]; circle maps, Arnold tongues, devil's staircases and torus breakdown in Arnold [1], Aronson et al. [2], Curry and Yorke [7], Jensen et al. [12, 13], Ostlund et al. [23], Ruelle and Takens [24]. A literature check using generic queries found no report of quasi-periodic training, Arnold tongues or rotation-number staircases for optimizers acting on losses; Hofmann [11] derives complex multipliers of a 2-cycle without rotation numbers, and tongues are documented in learning dynamics for games [22]. Every citation in this draft still needs checking at its primary source.

9 Reproducibility

Code, plans and compact receipts live in the project repository; bulky raw maps live in a separate data store referenced by SHA-256. Every figure is generated by a script in `paper/gd-route/figures/` that reads saved evidence and writes a receipt with input hashes, the generator's hash, output hashes, alt text and the values it displays. Every number in the text is a macro generated by `paper/gd-route/scripts/build_numbers.py`; `data/numbers.json` records each macro's source file, hash and computation, and `provenance.json` indexes all receipts. `paper/gd-route/scripts/verify.py` regenerates every figure and number and fails on any difference. The map study corresponds to the repository's EXP-028 plan and audit; its plan, receipts and audit are linked from `docs/EXP-028-results.md`.

Placeholder. Author contributions, acknowledgments, license choice and public code and data links are human decisions to be made before release.

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